

Iteration-denial

Cian Dorr

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1. Physicalistic reasons for rejecting Hypertolerance claims

Weak Microphysical Supervenience: Every true proposition is necessitated by some true microphysical proposition.

Strong Microphysical Supervenience: Every possibly-true proposition is necessitated by some possibly-true microphysical proposition.¹

- What's a "microphysical proposition"? Intuition: one that could be expressed in a mighty language with predicates for microphysical properties and relations and names for microphysical objects. Extremely challenging to turn this into a rigorous definition even if we are happy to help ourselves to notions like microphysical properties, relations, and objects.

Let a *specific* property be one that necessitates *being such that p* for some microphysical *p*.

In a *de re* Tolerance Argument where all the properties involved are specific, and some instantiated property that isn't instantiated by the relevant object *a* is *ancestrally close* to one instantiated by *a*, Hypertolerance conflicts with Weak Microphysical Supervenience. And if two of the properties ancestrally close to one instantiated by *a* are incompatible but possibly both instantiated, Hypertolerance conflicts with Strong Microphysical Supervenience.

- In the book we attempt to construct an argument like this, involving earrings.

2. Coarse Hypertolerance

Note that the Tolerance premises required for Tolerance Arguments where Hypertolerance conflicts with some version of Microphysical Supervenience are much further removed from everyday life than more familiar Tolerance premises. So there's room for a hybrid strategy: accept ordinary ("coarse") Tolerance claims and the corresponding Hypertolerance claims, but reject both Tolerance and Hypertolerance in the cases where Hypertolerance would conflict with Microphysical Supervenience.

- The worlds at which any given ordinary object is concrete are sprinkled like dust across a broad swath of modal space.

¹ Note: a completely different use of 'weak' and 'strong' is common in the literature on supervenience understood as a relation between sets of *properties*.

- Robustness worry: we think not only that each table *could* have been a little different, but that it would be hard to make something only a little different that *wasn't* that table.

3. Tolerance Puzzles and the denial of Metaphysical Iteration

For philosophers including Chandler, Salmon, and Robertson

Temporal Iteration and temporal Tolerance Puzzles

Williamson's puzzle of the unmade earrings

4. Ancestral necessity and Ancestral Microphysical Supervenience

It is ancestrally necessary that p : p instantiates every finite iteration of metaphysical necessity.

Let Ancestral Weak and Strong Microphysical Supervenience be the results of substituting ancestral for metaphysical necessity in Weak and Strong Microphysical Supervenience.

5. Arguing from higher-order identities to ancestral necessities

(1) It is ancestrally necessary that an object is in the Tri-State Area just in case it is in New York, New Jersey, or Connecticut.

(2) To be in the Tri-State Area is to be in New York, New Jersey, or Connecticut.

(2) has a natural reading as a *higher-order identity* claim. So understood, it implies (1) given LL and

(3) It is ancestrally necessary that an object is in the Tri-State Area just in case it is in the Tri-State Area.

Some other plausible identities in a similar vein:

(4) For m to be an oxygen molecule is for $a1$ and $a2$ to compose an oxygen molecule.

(5) Muhammed Ali's left fist and Muhammed Ali's left hand are such that: to be hit by the former is to be hit by the latter while it is clenched.

(6) Plato's beard is such that: to be identical to it is to be a beard of Plato.

A general picture suggested by these examples: "non-fundamental objects are linked to fundamental ones by a rich pattern of higher-order identities". This motivates Ancestral Strong Microphysical Supervenience (or something sufficiently like it).

- Objection: in the case of tables, pyramids, etc., there is nothing plausible we could write down analogous to (2,4–6).

- Response: (i) thanks to vagueness, nothing we could write down would be true-on-all-precisifications. (ii) anything that's even true on *some* precisifications would be very complex.

6. Structure

While identities like (2) and (4–6) are *prima facie* attractive, there are possible principled reasons for rejecting them. The most influential are based on certain generalizations of:

Structure $Fx = Gy \rightarrow (x = y \wedge F = G)$

Generalized Structure

- (i) $(\lambda x_1 \dots x_k. Rx_1 \dots x_k y_1 \dots y_n \neq \lambda x_1 \dots x_k. Sx_1 \dots x_k z_1 \dots z_m \text{ (where } m=n))$
- (ii) $\lambda x_1 \dots x_k. Rx_1 \dots x_k y_1 \dots y_n = \lambda x_1 \dots x_k. Sx_1 \dots x_k z_1 \dots z_n \rightarrow R=S \wedge y_1=z_1 \wedge \dots \wedge y_n=z_n.$

7. The Russell-Myhill theorem

Super-Weak Propositional Structure: $\forall F \forall G \forall p [(F(p) = G(p)) \rightarrow \forall q (F(q) \leftrightarrow G(q))]$

This says that: for any p , “applying to p ” generates a mapping from type $\langle\langle\rangle\rangle$ to type $\langle\rangle$ in which only coextensive type- $\langle\langle\rangle\rangle$ things are mapped to the same type- $\langle\rangle$ thing. In the light of Cantor’s theorem, the following result is thus not surprising:

Russell-Myhill Theorem: *Super-Weak Propositional Structure* is inconsistent in the background logic.

Proof: Choose some arbitrary p . Let $O =_{df} \lambda q^{\diamond}. \neg \forall F ((q = F(p)) \rightarrow F(q))$.

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|--|---------------------|
| (1) $O(Op) \leftrightarrow \neg \forall F (Op = Fp \rightarrow F(Op))$ | Extensional β |
| (2) $\forall F (Op \equiv Fp \rightarrow F(Op)) \rightarrow (Op = Op \rightarrow O(Op))$ | UI |
| (3) $\neg O(Op) \rightarrow (Op = Op \rightarrow O(Op))$ | (1, 2) |
| (4) $Op = Op$ | Ref |
| (5) $\neg O(Op) \rightarrow O(Op)$ | (3, 4) |
| (6) $O(Op)$ | (5) |
| (7) $\neg \forall F (Op \equiv Fp \rightarrow F(Op))$ | (1, 6) |
| (8) $\neg \forall F (Op \equiv Fp \rightarrow (O(Op) \leftrightarrow F(Op)))$ | (6, 7) |
| (9) $\neg \forall F (Op \equiv Fp \rightarrow \forall q (Oq \leftrightarrow Fq))$ | (7, UI) |

Line (9) is the denial of Super-Weak Propositional Structure. Since p was arbitrary, we can conclude that for any p , there are some non-coextensive operators F and G such that $F(p) = G(p)$.

Let me attempt at a gloss in English. Define: q is a stone-thrower iff q attributes to p some property that q lacks.

- (1): The proposition that p is a stone thrower is a stone-thrower iff it attributes to p some property that it itself lacks.

- (2): If the proposition that p is a stone-thrower doesn't attribute to p any property that it itself lacks, then if it attributes being a stone-thrower to p , it is a stone-thrower.
- (3): If the proposition that p is a stone-thrower isn't a stone-thrower, then if it attributes being a stone-thrower to p , it is a stone thrower.
- (4): The proposition that p is a stone thrower attributes being a stone-thrower to p .
- (5): If the proposition that p is a stone-thrower isn't a stone-thrower, then it is a stone thrower.
- (6): The proposition that p is a stone-thrower is a stone-thrower.
- (7): There is some property that the proposition that p is a stone-thrower attributes to p , but deoesn't have itself.
- (8): There is some property that the proposition that p is a stone-thrower attributes to p , where it's not the case that it has it itself iff it's a stone-thrower.
- (9): There is some property that the proposition that p is a stone-thrower attributes to p , which is not coextensive with (and hence not identical to) being a stone-thrower.

8. Weakenings of Structure ruled out by Beta-Identity

Atomic Structure $Fx = Gy \rightarrow (F = G \wedge x = y)$ for any *objects* x and y .

Atomic Structure (and its polyadic generalization) is consistent (in our basic background logic). However it is inconsistent with

Beta-Identity $(\lambda v_1 \dots v_n. \phi)(a_1, \dots, a_n) = \phi[a_1 / v_1, \dots, a_n / v_n]$

Example: Beta-Identity implies $(\lambda x. Lxc)(c) = Lcc = (\lambda x. Lcx)(c)$

whereas Atomic Structure implies $(\lambda x. Lxc)(c) = (\lambda x. Lcx)(c) \rightarrow (\lambda x. Lxc) = (\lambda x. Lcx)$

- This also shows the inconsistency of Atomic Structure with various weakenings of Beta-Identity, such as *Non-Vacuous Beta Identity* (whose instances are the instances of Beta-Identity where each of $v_1, \dots, v_n$ has at least one free occurrence in ϕ).

9. Weakenings of Structure consistent with Beta-Identity

Qualitative Atomic Structure $(\text{Qual}(F) \wedge \text{Qual}(G) \wedge Fx = Gy) \rightarrow (F = G \wedge x = y)$

This is consistent with Non-Vacuous Beta-Identity.

Weak Qualitative Atomic Structure

$(\text{Qual}(F) \wedge \text{Qual}(G) \wedge \forall p (F \neq (\lambda x. p) \wedge G \neq (\lambda x. p))) \wedge Fx = Gy \rightarrow (F = G \wedge x = y)$

Finding a consistent polyadic generalization of [Weak] Qualitative Atomic Structure is tricky but possible. The following will do:

Generalized Qualitative Atomic Structure If R and S are qualitative, and objects $y_1, \dots, y_n$ are all distinct, and objects $z_1, \dots, z_m$ are all distinct, and $\lambda x_1 \dots x_k. R y_1 \dots y_n x_1 \dots x_k = \lambda x_1 \dots x_k. S z_1 \dots z_m x_1 \dots x_k$ then $m = n$ and for some permutation $\pi 1, \dots, \pi n$ of the numbers $1, \dots, n$: y_1 is $z_{\pi 1}$ and ... and y_n is $z_{\pi n}$ and S is $\lambda u_1 \dots u_n x_1 \dots x_k. R u_{\pi 1} \dots u_{\pi n} x_1 \dots x_k$.

10. Is special pleading for objects well-motivated?

Putative counterexamples to GQAS involving abstract objects:

- (1) For Obamapair to be a pair of people married to one another is for Barack Obama to be married to Michelle Obama.
- (2) For zero to number-the-children-of Wittgenstein is for Wittgenstein not to have any children.

Possible response: things like numbers and ordered pairs aren't objects. Quantification over them is "concealed higher order quantification", analogous to how we've been using words like 'property'.

Possibly reply: If this is tenable and well-motivated for abstract objects like numbers and ordered pairs, it's also tenable and well-motivated for non-fundamental objects like areas, molecules, fists, beards, tables, pyramids,

Some simple versions of this vision worth considering.

- The only real ('type- e ') objects are *particles*. Ordinary material objects are properties of particles—i.e. quantification over them is type- $\langle e \rangle$. For particle x to be part of object F is for it to be the case that Fx .
- The only real ('type- e ') objects are spacetime points. Ordinary material objects are properties of spacetime points—i.e. quantification over them is type- $\langle e \rangle$. For object F to be located at point x is for it to be the case that Fx .